

Integrating Supervised and Self-Supervised Models for an Enhanced PCOS Detection: A Data-Driven Approach with Machine Learning Perspective

Tasmia Tahmida Jidney
CCDS
Independent University, Bangladesh
Dhaka, Bangladesh
tasmiaahmida.8@gmail.com

Syed Tangim Pasha
CCDS
Independent University, Bangladesh
Dhaka, Bangladesh
syed15-8631sets@iub.edu.bd

Mehedi Hassan
Dept. of Information Technology
Georgia Southern University
GA, USA
mehedihassan@georgiasouthern.edu

Ashraful Islam
CCDS
Independent University, Bangladesh
Dhaka, Bangladesh
ashraful@iub.edu.bd

M. Ashraful Amin
CCDS
Independent University, Bangladesh
Dhaka, Bangladesh
aminmdashraful@iub.edu.bd

Abstract—Polycystic Ovary Syndrome (PCOS) is a common ovarian dysfunction that leads to various difficulties, including missed or irregular menstrual periods, acne, depression, mood swings, and excessive facial hair. Due to the wide range of symptoms, diagnosing PCOS can be challenging. In addition, the process can be costly, and there is a high rate of false positives in PCOS diagnoses. Machine Learning (ML) can aid both patients and clinicians in diagnosing PCOS by using labeled data to analyze patient history effectively. However, the Self-Supervised model uses labeled and unlabeled data to understand new patient cases better. Incorporating unlabeled information during model training has become increasingly reliable for identifying a new patient case study. In this study, we employed the publicly available KagglePCOS tabular dataset to evaluate various modeling approaches. Our findings indicated that the Support Vector Classifier (SVC) model achieved a commendable accuracy of 93.8%. In comparison, the Gaussian Naive Bayes (GNB) model exhibited notably lower performance, with an accuracy of 54.1%. Among the Self-Supervised models, the Autoencoder showcased the most effective balance between accuracy and recall, achieving an accuracy of 82.7%. Although other models, such as Simple Contrastive Learning of Representations (SimCLR) and Bootstrap Your Own Latent (BYOL), demonstrated lower F1 scores and accuracy, they highlighted the potential of Self-Supervised methods in identifying new cases within unlabeled datasets. While traditional Supervised models remain highly effective, Self-Supervised learning offers a promising direction, especially for diagnosing cases where labeled data is unavailable.

Index Terms—Polycystic Ovary Syndrome, KagglePCOS Dataset, Supervised Learning, Self-Supervised Learning, Autoencoder, SimCLR, BYOL.

I. INTRODUCTION

According to the World Health Organization, 8–13% of women are affected by Polycystic Ovary Syndrome (PCOS) in reproductive age [1]. This syndrome is a common gynecological disorder in women in their reproductive age [2]. The main complexity of the diagnosis of PCOS is that the

symptoms are not constant and can differ for different women, which makes the recognition and prevention of the disease more complicated [3]. Most women are unaware of this disease until they face difficulties in pregnancy or a long-term absence of a period [4]. There are some common symptoms, *e.g.*, irregular menstrual cycles, hair growth on the face, among other symptoms, diabetes, and endocrine disorders, *e.g.*, infertility, metabolic abnormalities, and worsening [5]. Because these symptoms vary, establishing a consistent diagnostic method for PCOS is challenging. This shows the importance of getting more accurate, high-tech diagnostic tools to deal with a wider range of diseases more effectively and widely.

Machine Learning (ML) and Deep Learning (DL) can play a transformative role. These models can analyze large datasets, including a patient's medical history, hormone production patterns, and parts of ultrasound images, to find patterns that doctors might miss. This is because humans need time to consider every part of a task. Furthermore, ML can also learn from labeled and unlabeled data and thus make it smarter over time while simultaneously being capable of handling more complex cases. Recently, ML models have shown promising results in the medical sector. Researchers extensively use various Supervised learning models to diagnose any medical condition. However, such methods depend on huge labeled data, which may not always be available, especially in medical-related cases [6]. Although studies have demonstrated the effectiveness of Supervised learning, utilizing unlabeled data can present challenges. Recently, researchers have created Self-Supervised models to address this issue, enabling the pre-training of neural networks and enhancing their reliability. The results for advanced architectures such as Autoencoder, Bootstrap Your Own Latent (BYOL), and Simple Contrastive Learning of Representations (SimCLR) suggest great promise.

These advanced architectures have the potential to uncover concise representations of data that are hidden within large datasets without necessitating extensive manual labeling. We will use these Self-Supervised techniques with Supervised data if the need arises again. They consistently outperform traditional Supervised techniques, particularly in scenarios with limited annotations. We have used a publicly available benchmark tabular dataset called “KagglePCOS,” which contains 541 rows and 45 columns. The contributions of this paper are written below:

- This paper aims to analyze different aspects and prospects of Self-Supervised learning models for PCOS detection.
- We compare the results of different Supervised models between Logistic Regression (LR), Random Forest Classifier (RFC), Decision Tree Classifier (DTC), Gradient Boosting Classifier (GBC), Support Vector Classifier (SVC), K-Nearest Neighbors (KNN), Adaptive Boosting (AdaBoost) Classifier, Gaussian Naïve Bayes (GNB), Quadratic Discriminant Analysis (QDA), Ridge Classifier (RC), Passive Aggressive Classifier (PAC), and evaluate efficiency in PCOS detection.
- We analyze Self-Supervised models such as Autoencoder, BYOL, and SimCLR and evaluate their efficiency in PCOS detection.
- The results compare Supervised and Self-Supervised models for disease detection.
- This work also emphasizes the role of independent and self-learning procedures in diagnosing medical conditions.

The rest of the paper is organized as follows: Section II presents the literature review on PCOS detection. In Section III, we describe our dataset, explain the data preprocessing steps, and provide details of the experimental setup. Section IV compares various Supervised and Self-Supervised models for PCOS detection. Finally, we conclude our study and discuss future work in Section V.

II. LITERATURE REVIEW

A. Traditional Diagnostic Approaches for PCOS

The research paper by Kathleen et al. [7] published an in-depth overview of PCOS, focusing on its complexity of diagnosis and treatment challenges. It highlights the 2018 international guidelines for managing PCOS and emphasizes the importance of lifestyle and food habit changes, as well as suggesting medical treatments, *e.g.*, combined oral contraceptives and metformin. This paper points out the reasons for infertility, metabolic issues, slow weight loss, and menstrual irregularities as lifestyle interventions. In consequence, another study from Copp et al. [8] explores the long-term health impacts of PCOS in women living in Western Australia. This paper investigates hospital admission records over 15 years and finds an interesting point that women with PCOS had significantly higher rates of hospitalizations for various health conditions, *e.g.*, diabetes, obesity, cardiovascular diseases, mental health disorders, and reproductive complications, including

infertility and miscarriage. Women with PCOS often require fertility treatments, *e.g.*, IVF. PCOS also impacts long-term mental health and psychological issues. These women have less control over stress and also suffer from anxiety issues. The paper by Tracy et al. [9] discusses the diagnosis and treatment of PCOS, which affects around 7% of reproductive-aged women. The Rotterdam criteria for diagnosis link the condition to insulin resistance and metabolic issues, *e.g.*, type 2 diabetes. Treatment varies based on symptoms, with weight loss, hormonal contraceptives, and medications, *e.g.*, clomiphene and metformin, being commonly used to manage infertility, menstrual irregularities, and metabolic concerns. In separate research, Blazej et al. [10] examined the difficulties associated with diagnosing and treating PCOS in adolescents, where symptoms, *e.g.*, irregular menstruation and hyperandrogenism, overlap with normal puberty. It emphasizes careful diagnostic criteria and highlights the metabolic and psychological risks, including insulin resistance and mood disorders. They suggest initial treatment as lifestyle changes, followed by medications, *e.g.*, contraceptives and COPP [8].

B. ML approaches for PCOS diagnosis

Ashwini et al. [11] explore ML models, *e.g.*, Support Vector Machine (SVM) and Random Forest (RF) applications for PCOS diagnosis. It highlights the potential of Artificial Intelligence (AI) to provide faster and more accurate diagnoses than traditional methods. However, the dataset of this paper is too small to consider as any benchmark research. Another study by Chan et al. [12] demonstrates the effectiveness of the Gradient Boosted Decision Tree (GBDT) algorithm in improving the accuracy of PCOS detection by analyzing complex clinical data. Although their approach is very promising and lacks enough data, we also fail to consider it the art of the state. Similarly, Vaidehi et al. [13] proposed the system “PCOcare” for detecting PCOS in the primary stage. By using the Chi-Square method for feature selection, they achieved 90.9% using the RF Classifier model. The proposed PCOcare system helps in early diagnosis, better health management, and reduced health risks for both patients. Satish et al. [14] explore using ML techniques to predict PCOS using a dataset from Kaggle. Models like RF achieve 93.12% accuracy in RapidMiner. Feature selection methods helped identify key factors for effective prediction, *e.g.*, BMI and follicle count.

C. DL Approaches for PCOS diagnosis

There is also some significant research using DL for PCOS diagnosis. Abrar et al. [15] did an ultrasound image and clinical data-based research and proposed a DL fusion model. They achieved 84.81% accuracy using ultrasound images. Later, they implemented a fusion approach and improved over 82.46% accuracy with MobileNet. This system is supposed to assist radiologists in diagnosing PCOS better and more efficiently. The paper by Sowmiya et al. [16] introduces F-Net, a DL model for diagnosing PCOS using ultrasound images only. The model uses YOLOv8 for follicle detection and the Fuzzy C-Means (FCM) clustering technique for segmentation.

F-Net achieved the highest accuracy, 95%, and 97.5%, on two datasets, outperforming other ML and DL models, which makes it an effective tool for automated PCOS diagnosis.

III. METHODOLOGY

The entire methodology in this work is illustrated in Fig. 1.

A. Dataset Description

Our dataset contains medical and lifestyle information for 541 patients, focusing on PCOS [17]. It includes 45 columns and 541 rows. This dataset has demographic details for patients. This information was collected from 10 different hospitals across India. The dataset is provided in two separate CSV files, one containing the features and the other containing the target values (PCOS diagnosis). Table I presents the dataset description and the preprocessing steps used in our research.

B. Dataset Preprocessing

After merging two CSV files (features, target), unnecessary columns (*e.g.*, unnamed columns) were removed to refine the dataset. Then, we checked for null values and replaced them with zero to ensure a clean dataset. Upon analyzing the distribution of the target variable PCOS (Y/N), it was clear that the dataset was imbalanced. To address this, we applied the ADASYN (Adaptive Synthetic Sampling) method. This method generates synthetic data points for the minority class, which balances the dataset. After resampling, the dataset contained 729 rows with 43 feature columns and 1 target column. We standardized the features using StandardScaler to

TABLE I
DATASET DESCRIPTION AND PREPROCESSING STEPS.

Dataset Details	Description
Source	10 hospitals in Kerala, India [17]
No. of Patients	541
Features (Columns)	45 columns: demographics, vitals, reproductive health, lifestyle factors
File Format	Two CSV files (features, target)
Merging Method	Merged six common columns, <i>e.g.</i> , SI. No., Patient File No., PCOS (Y/N), I beta-HCG (mIU/mL), II beta-HCG (mIU/mL), AMH (ng/mL)
Data Cleaning	Removed unnecessary columns, replaced missing values with zero
Target Variable	PCOS (Y/N)
Imbalanced Data	Handled via ADASYN (Adaptive Synthetic Sampling)
Final Dataset Size	729 rows, 43 features, 1 target column
Train/Test Split	80% Training, 20% Testing
Feature Scaling	Standardized using StandardScaler

normalize the data before applying ML models. We have split the dataset into 80:20 for training and testing.

C. Model Selection

In this research, we utilized medical data to train 11 Supervised and 3 Self-Supervised models. Among Supervised models, *i.e.*, LR, RFC, DTC, GBC, SVC, KNN, AdaBoost Classifier, GNB, QDA, RC, and PAC, as well as Self-Supervised models, *i.e.*, Autoencoder, SimCLR, and BYOL on our target variable PCOS (Y/N). While traditional Supervised models work on labeled data, Self-Supervised models work without any data labeling. To train our self-supervised models, we removed the target column. These models learn data by reconstructing only the input features (X) without using the labels (y).

D. Evaluation Metrics

Evaluation metrics are important for assessing a model's performance. We used recall, precision, accuracy, and F1-score metrics to evaluate our model's performance in this experiment.

1) *Accuracy*: The ratio of correctly predicted instances to all instances is called Accuracy, and the formula is:

$$\text{Accuracy} = \frac{TP + TN}{TP + FN + TN + FP} \quad (1)$$

2) *Precision*: The proportion of correctly predicted positive instances to all expected positive instances is called Precision, and the formula is:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (2)$$

3) *Recall*: The proportion of correctly predicted positive instances to the total number of actual positive instances is called Recall, and the formula is:

$$\text{Recall} = \frac{TP}{TP + FN} \quad (3)$$

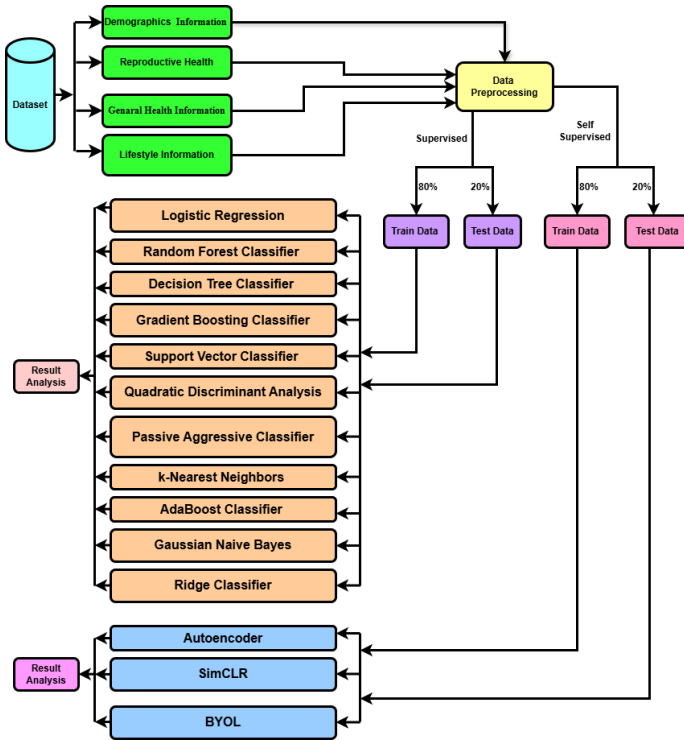


Fig. 1. Our Proposed Methodology Framework

4) *F1-Score*: The harmonic mean of Recall and Precision is called the F1-score, and the formula is:

$$F1\text{-score} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4)$$

IV. RESULTS ANALYSIS

A. Proposed Supervised Learning Models Performance Analysis

Traditional ML models, *e.g.*, LR, RFC, DTC, GBC, SVC, QDA, PAC, KNN, RC, AdaBoost, and GNB, were evaluated on a binary classification task. These ML models can handle complex features of the clinical datasets and diagnose any medical condition. Fig. 2 shows the ROC curves of our proposed Supervised models. In addition, Table II demonstrates the performance metrics of our proposed Supervised models. SVC and LR, in particular, perform better in PCOS diagnostics. SVC leads with an AUC of 0.9861, closely followed by RF at 0.985. These models achieve balanced results, which makes them the most reliable for this task. LR and KNN also performed well, with AUCs exceeding 0.96. However, models

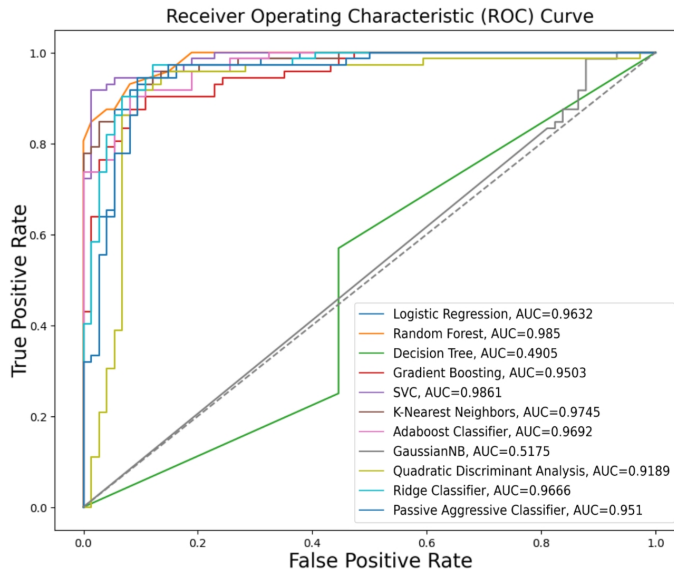


Fig. 2. Comparison of ROC Curves for Proposed Supervised Models.

TABLE II

COMPARISON OF PERFORMANCE METRICS FOR SUPERVISED MODELS IN PCOS DIAGNOSIS

Model Name	Accuracy (%)	Precision	Recall	F1-score
Logistic Regression	91.1	92	90	91
Random Forest Classifier	83.56	93	78	84
Decision Tree Classifier	81.50	91	77	82
Gradient Boosting Classifier	89.73	85	94	90
Support Vector Classifier	93.84	93	94	94
Quadratic Discriminant Analysis	89.72	85	94	89
Passive Aggressive Classifier	90.41	91	89	90
K-Nearest Neighbors	89.04	84	95	89
Ridge Classifier	91.09	90	91	91
AdaBoost Classifier	66.44	81	63	68
Gaussian Naive Bayes	54.11	11	88	65

like the DTC and GNB perform poorly, especially the DTC, with an AUC of 0.4905, indicating challenges in accurately diagnosing PCOS cases.

B. Proposed Self-Supervised Learning Models Performance Analysis

Self-Supervised models, *e.g.*, Autoencoder, SimCLR, and BYOL, show varying performance levels when diagnosing PCOS. With an AUC of 0.87, the Autoencoder outperforms the others, effectively balancing true positive and false positive rates. It demonstrates strong sensitivity, making it the best self-supervised model in this analysis. SimCLR, with an AUC of 0.82, performs moderately but falls short of the Autoencoder, while BYOL, with an AUC of 0.78, has the weakest performance, and these ROC curves are shown in Fig. 3. Moreover, Table III demonstrates the performance metrics of our proposed Self-Supervised models. Here, SimCLR struggles to differentiate between true positive and false negative cases effectively in our dataset. Overall, the Self-Supervised models show potential but require further refinement.

C. Comparative Performance Analysis of Supervised and Self-Supervised Models

In this study, Supervised models performed better than Self-Supervised models. The Supervised models with the best accuracy were SVC and LR. They consistently got higher accuracy and AUC values than the proposed Self-Supervised models. This is because labeled data is available, which lets Supervised models learn directly how to connect input features to the output of the PCOS diagnosis. We can also observe that

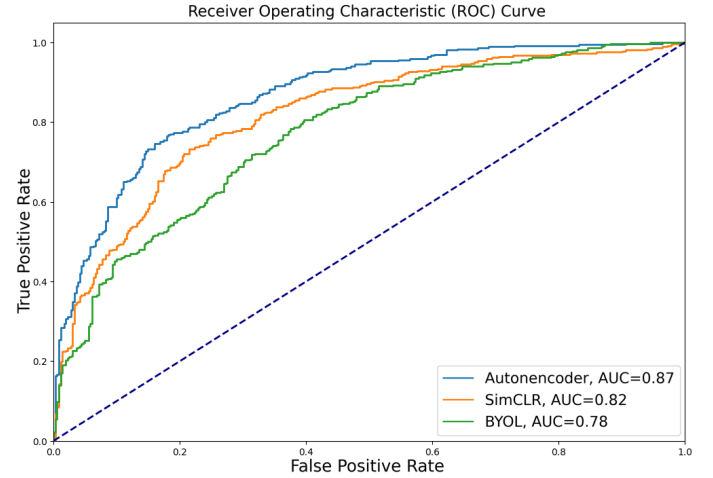


Fig. 3. Comparison of ROC Curves for Proposed Self-Supervised Models.

TABLE III

COMPARISON OF PERFORMANCE METRICS FOR SELF-SUPERVISED MODELS IN PCOS DIAGNOSIS

Model Name	Accuracy (%)	Precision	Recall	F1-score
Autoencoder	82.7	82.4	83.3	82.9
SimCLR	75.8	76.7	74.6	75.6
BYOL	71.9	73.1	69.7	71.3

the Supervised models, especially DTC and GNB, produced more false negatives. The complex, non-linear relationships in the data made it hard for these models to work. For example, they got PCOS cases that were borderline or not typically wrong. On the other hand, SVC and RF managed to minimize false negatives, which makes them reliable for better diagnosis.

In contrast, Self-Supervised models, which use unlabeled data to learn about it, give us lower accuracy and recall. This suggests their optimization for medical diagnosis tasks, *e.g.*, PCOS detection, remains incomplete. However, large amounts of labeled data are unavailable in the medical sector, so we can't overlook Self-Supervised models. The Autoencoder, in particular, demonstrated strong sensitivity, suggesting that Self-Supervised approaches could complement Supervised models in future hybrid frameworks. In Self-Supervised models, the Autoencoder performed better at capturing the intricate patterns within the dataset, but models like SimCLR and BYOL faced difficulties in differentiating between subtle variations in hormone levels and ultrasound readings, leading to a slightly higher rate of misclassification.

D. Threat to Validity

The primary threat to validity was the size and availability of the dataset. Collecting medical data is expensive. For more accurate results, we should work with larger and more diverse datasets. However, this dataset has been collected from a single region. Including data from multiple areas could lead to more robust conclusions. We employed various ML and DL models, particularly advanced Self-Supervised models, including Autoencoders, SimCLR, and BYOL, without tuning hyperparameters. Optimizing hyperparameters and exploring hybrid models combining Supervised and Self-Supervised approaches could yield more fruitful results.

V. CONCLUSION AND FUTURE WORK

Diagnosing PCOS remains a complex task. ML and DL approaches hold promise in improving diagnostic accuracy nowadays. While ML models, *e.g.*, SVC and LR, perform well, the AdaBoost classifier and GNB show weaker performance. In Self-Supervised models, Autoencoder performs the best, while SimCLR and BYOL do not achieve promising results. Among all models, SVC achieved the highest accuracy. The main reason for the lower accuracy of the Self-Supervised model is the shortage and complexity of the dataset. However, we need a larger and more diverse dataset from all over the world to train our model and significantly improve its accuracy. Improved model performance can ensure that healthcare professionals and people can trust and relay the model's outputs. Enhancing Self-Supervised and Supervised models, especially through advanced architectures, *e.g.*, Transformers, could improve their diagnostic capabilities. For real-world applications, it is important to explore hybrid models that combine learning methods and make them work better in clinical settings, emphasizing interpretability. Finally, ensuring ethical and privacy compliance through techniques like Feder-

ated Learning will help develop scalable, trustworthy solutions for PCOS diagnosis.

REFERENCES

- [1] World Health Organization, "Polycystic ovary syndrome: Fact sheet," 2023. Accessed: 2024-10-05.
- [2] H. Islam, J. Masud, Y. N. Islam, and F. K. M. Haque, "An update on polycystic ovary syndrome: A review of the current state of knowledge in diagnosis, genetic etiology, and emerging treatment options," *Women's Health*, vol. 18, p. 17455057221117966, 2022.
- [3] A. E. Joham, T. Piltonen, M. E. Lujan, S. Kiconco, and C. T. Tay, "Challenges in diagnosis and understanding of natural history of polycystic ovary syndrome," *Clinical endocrinology*, vol. 97, no. 2, pp. 165–173, 2022.
- [4] Y. V. Louwers and J. S. Laven, "Characteristics of polycystic ovary syndrome throughout life," *Therapeutic Advances in Reproductive Health*, vol. 14, p. 2633494120911038, 2020.
- [5] A. Chaudhuri, "Polycystic ovary syndrome: Causes, symptoms, pathophysiology, and remedies," *Obesity Medicine*, vol. 39, p. 100480, 2023.
- [6] J. Egger, C. Gsaxner, A. Pepe, K. L. Pomykala, F. Jonske, M. Kurz, J. Li, and J. Kleesiek, "Medical deep learning—a systematic meta-review," *Computer methods and programs in biomedicine*, vol. 221, p. 106874, 2022.
- [7] K. M. Hoeger, A. Dokras, and T. Piltonen, "Update on pcpos: consequences, challenges, and guiding treatment," *The Journal of Clinical Endocrinology & Metabolism*, vol. 106, no. 3, pp. e1071–e1083, 2021.
- [8] T. Copp, D. Muscat, J. Hersch, K. McCaffery, J. Doust, B. Mol, A. Dokras, and J. Jansen, "Clinicians' perspectives on diagnosing polycystic ovary syndrome in australia: a qualitative study," *Human Reproduction*, vol. 35, no. 3, pp. 660–668, 2020.
- [9] T. N. Harrison and R. J. Chang, "Ovarian response to follicle-stimulating hormone in women with polycystic ovary syndrome is diminished compared to ovulatory controls," *Clinical endocrinology*, vol. 97, no. 3, pp. 310–318, 2022.
- [10] B. Meczekalski, O. Niwczyk, A. Kostrzak, M. Maciejewska-Jeske, G. Bala, and A. Szeliga, "Pcos in adolescents—ongoing riddles in diagnosis and treatment," *Journal of Clinical Medicine*, vol. 12, no. 3, 2023.
- [11] A. Kodipalli and S. Devi, "Prediction of pcpos and mental health using fuzzy inference and svm," *Frontiers in public health*, vol. 9, p. 789569, 2021.
- [12] J. L. Chan, K. Moran, and K. Vanhise, "Getting a grip on polycystic ovary syndrome," 2023.
- [13] V. Mendpara, N. R. Ganampet, S. Afroze, S. Gnanasekaran, S. Sahu, R. Atikah, S. Rajpal, P. Raghavan, K. Ajith, M. P. Parmar, *et al.*, "Long-term effects of treatment modalities on metabolic health, fertility outcomes, and cardiovascular risks in lean women with polycystic ovary syndrome (pcpos): A systematic review," 2023.
- [14] S. C. Nandipati and C. X. Ying, "Polycystic ovarian syndrome (pcpos) classification and feature selection by machine learning techniques," *Applied Mathematics and Computational Intelligence (AMCI)*, vol. 9, pp. 65–74, 2020.
- [15] A. Alamoudi, I. U. Khan, N. Aslam, N. Alqahtani, H. S. Alsaif, O. Al Dandan, M. Al Gadeeb, and R. Al Bahrani, "A deep learning fusion approach to diagnosis the polycystic ovary syndrome (pcos)," *Applied Computational Intelligence and Soft Computing*, vol. 2023, no. 1, p. 9686697, 2023.
- [16] S. Umapathy, O. Alhajlah, F. Almutairi, S. Aslam, and A. RK, "F-net: Follicles net an efficient tool for the diagnosis of polycystic ovarian syndrome using deep learning techniques," *Plos one*, vol. 19, no. 8, p. e0307571, 2024.
- [17] P. Kottarathil, "Polycystic ovary syndrome (pcpos)," 2020. Accessed: 2024-10-06.